



Effect of freestream noise and boundary-layer transition on hypersonic shock-wave/boundary-layer interaction unsteadiness

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Abstract

The low-frequency shock and separation unsteadiness resulting from a three-dimensional transitional shock-wave/boundary-layer interaction is investigated for different freestream noise levels and incoming boundary-layer states in a quiet Mach-6 wind tunnel. In low-disturbance flow, the cone-slice-ramp geometry is studied with and without upstream boundary-layer trips, resulting in upstream-transitional and interaction-transitional SWBLI, respectively. When applied, the trips excite the second-mode instability, and the unsteady separation motion is studied for a range of second-mode magnitudes in the transitional upstream boundary layer. The effect of freestream noise is studied by operating the wind tunnel in both its low-disturbance and conventional-noise configurations. Unsteady separation motion is tracked by identifying the shear-layer location in schlieren visualization videos; pressure fluctuations are measured in the incoming boundary layer and inside the separation bubble. Exciting the second-mode instability in the boundary layer in quiet flow initially increases the magnitude of the unsteadiness and alters its frequency content by amplifying oscillations at $Sr_L = 0.025\text{--}0.05$; however, as second-mode amplitude increases and separation length decreases, this trend reverses and the unsteadiness is suppressed. The comparison of upstream-transitional SWBLI in quiet and noisy flows suggests that the pressure fluctuations in the noisy freestream are interacting with the mechanism responsible for SWBLI and significantly amplifying the unsteadiness over all frequencies.

Keywords Shock-wave/boundary-layer interaction · Separation bubble unsteadiness · Wind-tunnel noise · Hypersonic ground testing

1 Introduction

The low-frequency shock and separation unsteadiness resulting from shock-wave/boundary-layer interactions (SWBLIs) has been the topic of frequent experimental and computational investigation [1–3]. The origins of this unsteadiness

have often been studied and debated, with specific interest and conflicting results around its sensitivity to external disturbances [1–5]. In 1975, Plotkin [6] developed a mathematical model—which has since been supported experimentally—in which the unsteadiness in the shock and separation bubble is initiated by velocity fluctuations in the incoming boundary layer [1, 7]. Experimental and computational analyses of separation at a supersonic compression corner have both found a causal relationship between the low-frequency separation unsteadiness and velocity fluctuations in the incoming turbulent boundary layer [8, 9].

While some researchers have found evidence linking boundary-layer velocity fluctuations to the SWBLI-induced unsteadiness, others have proposed alternate mechanisms to explain the behavior. In their direct numerical simulation (DNS) of a supersonic compression corner, Wu and Martin [10] observed that the SBWLI-induced unsteadiness was correlated with and preceded by motion at the reattachment point of the separation bubble; from this finding, they proposed that the unsteady system was the result of

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a feedback loop between the separation bubble, shock, and shear layer. Piponniau et al. [11] introduced a mathematical model in which the unsteadiness is caused by a cycle of fluid entrainment through the shear layer and mass recovery via shear-layer flapping. Priebe and Martin [12] performed DNS of a turbulent SWBLI and observed a low-frequency modulation of turbulent structures in the shear layer; they then theorized that the corresponding modulation of fluid entrainment rate was a possible origin of the unsteadiness. They proposed that the SWBLI-induced unsteadiness was the result of a downstream global instability in the separation region, with the upstream boundary layer having a weak influence over fully separated flows. In an attempt to reconcile the various theories, Clemens and Narayanaswamy [2] proposed that the shock, shear layer, and separation bubble constitute a forced dynamical system, which responds to external disturbances from both the upstream boundary layer and downstream shear layer, thereby producing the SWBLI-induced unsteadiness in question.

Much of the existing literature on SWBLI unsteadiness has focused on turbulent interactions, with relatively few studies on transitional interactions [1, 3, 13–16]. Threadgill et al. [17] studied a Mach-4 interaction-transitional SWBLI upstream of a finite-span compression corner and characterized the unsteadiness using high-speed schlieren imaging. They observed low-frequency unsteadiness at a Strouhal number ($Sr_L = fL_{\text{bub}}/U_\infty$) consistent with turbulent SWBLI unsteadiness ($Sr_L \approx 0.025$), with evidence of a downstream mechanism driving the unsteadiness. Threadgill et al. also clarified the distinction between *upstream-transitional* and *interaction-transitional* SWBLI, with intermittency measured in the upstream boundary layer for the former but not the latter. Combs et al. [18] investigated the dynamics of a cylinder-induced upstream-transitional SWBLI in a supersonic wind tunnel using high-speed schlieren imaging; they concluded that the low-frequency unsteadiness was driven by a downstream instability in the separation bubble. Despite being a 3D interaction, the SWBLI exhibited unsteadiness at Strouhal numbers similar to those associated with 2D interactions ($Sr_L = 0.04$ – 0.08) when the length scale was based on centerline interaction size. From 2D DNS of a shock impingement on a laminar boundary layer, Sansica et al. [19] concluded that the low-frequency unsteadiness was primarily influenced by mechanisms inside separation. They found that increasing separation bubble size caused greater unsteadiness and a decrease in the influence of the upstream boundary layer. Additional 3D DNS by Sansica et al. [20] applied forcing only at frequencies higher than those of the unsteadiness, resulting in transition near reattachment. They postulated that the low-frequency unsteadiness ($Sr_L = 0.04$) arose from the turbulent breakdown around reattachment, which caused upstream traveling disturbances.

The vast majority of experimental studies on SWBLI-induced unsteadiness have been performed in wind tunnels with conventional freestream noise levels [1, 2, 21–23]. Low-disturbance—or quiet—wind tunnels contain freestream pressure fluctuation levels that are one or more orders of magnitude below what is typical of other ground test facilities [24]. Weiss and Chokani [25] studied the effects of freestream noise on separation unsteadiness using a compression ramp in a low-disturbance supersonic wind tunnel with a turbulent boundary layer. Weiss and Chokani concluded that the unsteadiness was independent of tunnel noise; however, in these experiments, the low-disturbance test core ended upstream of the separation shock foot, so while most of the incoming boundary layer was subjected to lower freestream noise levels, the interaction region was not. Leidy et al. [26] studied a transitional 3D SWBLI in a conventional supersonic wind tunnel that saw a threefold increase in freestream noise at a known Reynolds number threshold; the higher noise level impacted trends in separation size, surface heating, and pressure fluctuation spectra. They emphasized the need for studies of SWBLI in low-disturbance tunnels, especially since the impacts of freestream noise have rarely been documented.

While the majority of studies on SWBLI dynamics are performed on 2D interactions, the understanding of three-dimensional effects is critical for real-world applications [3]. The cone-slice-ramp geometry was developed to generate a complex 3D interaction resembling the effects of a deflected control surface [27, 28]. This geometry features a finite-span compression ramp mounted on the flat surface formed by taking a slice parallel to the axis of a straight cone; however, the flow features of the resulting 3D interaction differ significantly from those of a 2D SWBLI caused by a compression ramp. Experimental [27, 29] and computational [30, 31] studies of the cone-slice-ramp have observed a three-dimensional separated-flow region with curved separation and reattachment fronts; the separation bubble has no single length scale and is longest along the symmetry plane of the geometry. Simulations of the cone-slice-ramp by Thome et al. [30] also revealed a pair of counter-rotating vortices inside of the separation region, with flow spillage occurring around the sides of the ramp. In experimental studies [32–34] of an interaction-transitional 3D SWBLI on the cone-slice-ramp in quiet Mach-6 flow, the separation bubble and shock exhibited a low-frequency unsteadiness with the separation point fixed at the leading edge of the slice. According to the definitions specified in [35], this unsteadiness is classified as a *flapping* motion as opposed to the *breathing* motion more frequently observed in SWBLI studies. While flapping is sometimes used to describe the shear-layer oscillations near reattachment, in this case it refers to the large-scale motion of both the shear layer and shock.

Despite extensive research on SWBLI unsteadiness showing its sensitivity to external disturbances, most experiments are conducted in facilities with non-negligible freestream noise, and as a result, it remains unclear whether the experimentally observed low-frequency unsteadiness is influenced by pressure fluctuations in the freestream [1, 2, 36]. The present work seeks to isolate and examine the effects of wind-tunnel noise on the unsteady behavior of a transitional 3D SWBLI. In addition, the SWBLI system's response to varying pressure fluctuation levels in a transitional upstream boundary layer containing the second-mode instability is examined in quiet Mach-6 flow to determine the influence of transition on the unsteadiness.

2 Equipment and methods

2.1 Boeing/AFOSR Mach-6 Quiet Tunnel

Experiments are performed in the Boeing/AFOSR Mach-6 Quiet Tunnel (BAM6QT) at Purdue University, at a unit Reynolds number of $9.5 \pm 0.4 \times 10^6/\text{m}$. The BAM6QT facility is able to produce freestream flow with pitot pressure fluctuation levels on the order of 0.01%, compared to the 0.5 to 3% present in other hypersonic wind tunnels [37]. This low freestream noise level is possible because of a mirror-like finish on the nozzle, a gradual contour in the diverging section, and boundary-layer suction at the throat through a bleed valve [24]. By disabling this boundary-layer suction, the facility can also be operated in its noisy configuration, with freestream pressure fluctuation levels of approximately 3% and a Mach-5.8 freestream [38]. Downstream of the test

section, a pair of metal diaphragms separates the pressurized region from the downstream vacuum chamber (see Fig. 1).

A test in the BAM6QT begins by rupturing these diaphragms, allowing hypersonic flow to develop. After tunnel startup lasting up to 0.7 s, steady quiet flow is maintained for approximately 1.5 s. All reported times within a test, t , are measured relative to the burst time of the diaphragms. During a test, expansion waves reflect down the length of the driver tube approximately every 200 ms, each causing a decrease in stagnation pressure [39]. These pressure drops allow the model to experience a range of Reynolds numbers throughout a single test, with Reynolds number decreasing as time increases. Stagnation pressure is measured using an ETL-79-HA-DC-190 Kulite pressure sensor installed in the nozzle wall just upstream of the contraction section. The tunnel is filled with heated air so that the stagnation temperature at the start of each test is approximately 433 K.

2.2 Cone-slice-ramp geometry

The cone-slice-ramp wind-tunnel model used in the experiments was first developed by McKiernan [34] as a modified version of the original Oberkampf geometry [27] with a longer ramp to ensure complete reattachment of the shear layer. It consists of a 7-degree half-angle cone with a slice taken parallel to the cone's axis; a 20-degree finite-span ramp is mounted on the flat surface created by the slice (see [34] for a full model description). The cone-slice-ramp model includes a removable boundary-layer-trip insert, which features a ray of 12 equally spaced cylinders with a height of 635 μm and a diameter of 560 μm ; they are located 50.8 mm downstream of the sharp nose tip and span ± 54 degrees with respect to the model's plane of sym-

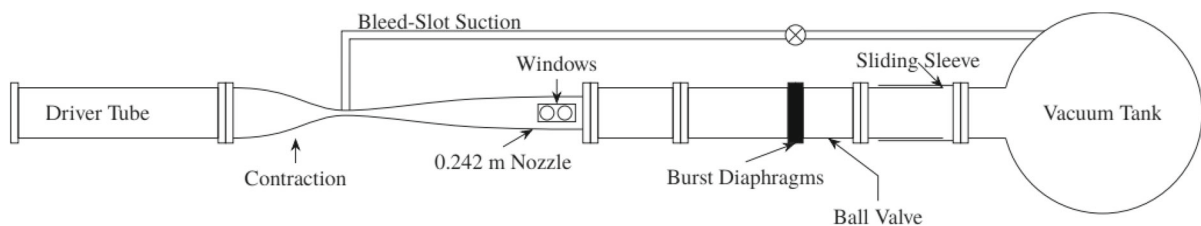
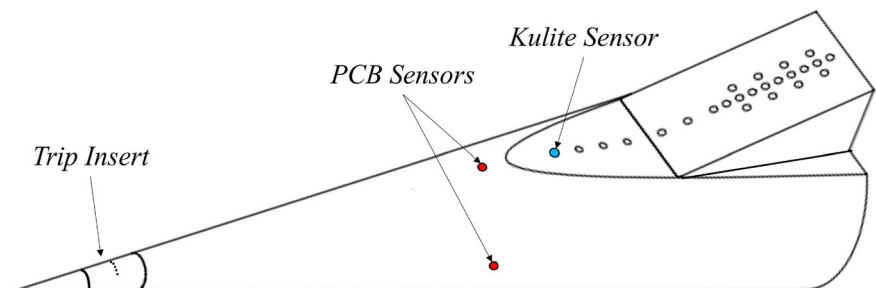


Fig. 1 Schematic of the Boeing/AFOSR Mach-6 Quiet Tunnel

Fig. 2 Schematic of cone-slice-ramp model with the trip insert and key sensor locations identified



metry. For the cases without boundary-layer trips, a smooth insert is installed in its place. The model is instrumented with PCB-132B38 and Kulite-XCE-062 pressure transducers to measure high- and low-frequency pressure fluctuations, respectively; their exact locations are shown in Fig. 2. The wind-tunnel model is tested at a nominally 0-degree angle of attack; four azimuthally spaced PCB sensors aid in angle-of-attack adjustments. The sensors are used to measure the peak frequency of Mack's second-mode instability around the cone in noisy flow, with mismatches in peak frequency indicating the presence of an off-axis angle. Corrections to the angle of attack can then be made accordingly via a precision sting adapter (see [40] for more information on angle-of-attack adjustments).

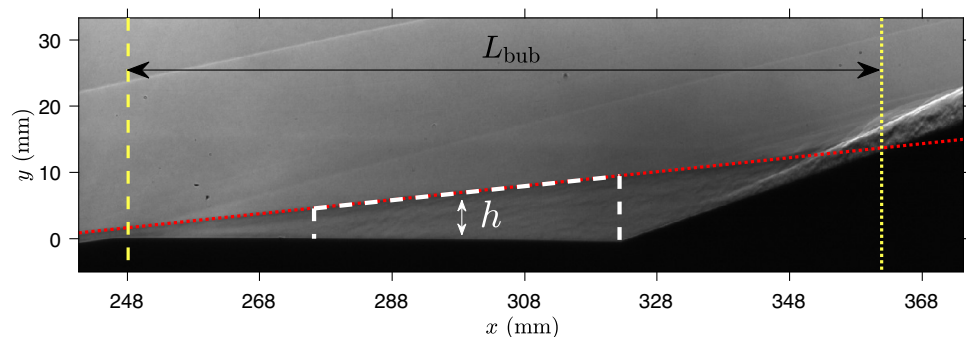
2.3 Schlieren system and processing techniques

A Z-type schlieren system is used to image the interaction region at 76,000 frames per second. The system uses a 10-ns duration light pulse produced by a Cavilux Smart laser, with a Phantom TMX high-speed camera recording each frame. The schlieren system has a resolution of 9.76 pix/mm; a more detailed description of the system can be found in [41]. The unsteady dynamics of the SWBLI are studied by tracking the motion of the separation bubble in the schlieren. An algorithm was developed to identify the edge of the shear layer over the last 45 mm of the slice, using a weighted average of pixel intensity around the location of highest intensity for each column of pixels (see Fig. 3). The slice coordinates are identified as the location of highest gradient in each column of pixels. To address issues caused by turbulent fluctuations in the shear layer negatively impacting the tracking accuracy for certain frames, a modification is implemented which applies the tracking method to the mean of the seven images surrounding the image in question. Because the rate of the fluctuations traveling along the shear layer is higher than the schlieren rate which is, in turn, much higher than the low-frequency separation oscillations, this mean image blurs out the offending turbulent structures without altering the measured low-frequency height fluctuations. The algorithm also identifies the ramp corner and slice locations in the schlieren

images, using these as the basis of a coordinate system that ensures continuity across all analyses. A linear fit is applied to the shear-layer coordinates, and the average distance between the shear layer and the slice is reported as the separation height, h , for each frame. The separation height is tracked in this manner for at least 3000 consecutive schlieren frames per analysis, so that the low-frequency content of the separation bubble motion is well-resolved. Because the separation front on this geometry does not translate, the fluctuations in bubble height are chosen as a representative measurement of the bubble's unsteadiness, which oscillates as if it is hinged at the leading edge of the slice. While the separation point is pinned at the beginning of the slice, the reattachment location is not as easily determined due to the three-dimensional reattachment behavior of this geometry. The separation bubble on this geometry reattaches earlier on the edges of the ramp, so the actual centerline reattachment point is obscured in the schlieren. In order to quantify separation bubble length, an estimated centerline reattachment point is obtained by extrapolating the visible shear layer to find its intersection point with the ramp. This process is performed on the average of 200 images to obtain the mean separation length, L_{bub} , at a specific time during a test. While 3D SWBLI cannot be characterized by a single length scale, the centerline separation length was chosen as a representative dimension in order to compare Strouhal numbers against existing literature. The calculation of Strouhal number based on a centerline length scale for 3D SWBLI has previously been applied to cylinder-induced separated flows [18, 42].

Schlieren images are also used to determine a mean upstream boundary-layer thickness, δ_u . The boundary-layer thickness is determined with a method similar to the one used in [43] for a mean of 200 images. The region of the boundary layer used in this analysis spans 3.5 mm, ending 1 mm upstream of the leading edge of the slice. The image gradient of the mean image is calculated, and a Weiner filter is applied to reduce the effects of noise from the camera sensor. For each column of pixels, the maximum gradients are used to determine the locations of the model and the edge of the boundary layer. As an additional step, a weighted average around the highest gradient is used to identify the boundary-

Fig. 3 Diagram showing how the separation tracking algorithm determines separation bubble height and length. Estimated centerline separation and reattachment points are indicated with the dashed and dotted yellow lines, respectively. The sample schlieren image is from the quiet-tripped case at $t \approx 2.20$ s



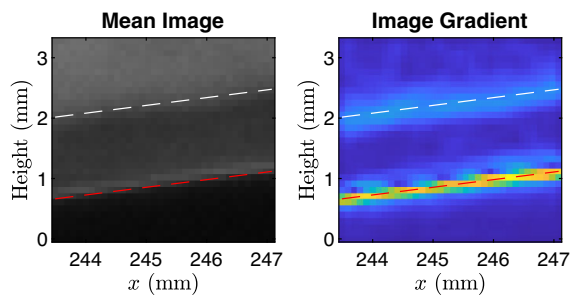


Fig. 4 Mean schlieren image and image gradient with dashed lines identifying the boundary-layer edge (white) and model edge (red). Sample images are from the noisy-smooth case

layer edge with sub-pixel resolution. Because the model's edge and boundary layer are tilted 7 degrees with respect to the camera, a linear fit is applied to each set of coordinates to ensure the edges are as well-defined as possible. An example of this process is shown in Fig. 4, which plots the identified edges of the model and boundary layer over the mean image and image gradient of the noisy-smooth case. The final upstream boundary-layer thickness is determined as the mean distance between these two lines, adjusted to account for the 7-degree tilt in the measurement frame so that the wall-normal thickness is recorded.

3 Results and discussion

3.1 Mean flow properties

It has been repeatedly observed [1, 13, 15, 44] that the dynamic behavior of SWBLI is highly dependent on upstream boundary-layer state. However, on the cone-slice-ramp, the noisy freestream condition causes a transitional upstream boundary layer, while the quiet freestream condition results in a laminar one at the same Reynolds number. Therefore, the effect of wind-tunnel noise alone cannot be studied without first altering the upstream boundary layer of the quiet flow case so that its mean properties are as close as possible to the one that forms naturally in noisy flow. This is achieved using the boundary-layer-trip insert described in Sect. 2.2. Experiments on a transitional 3D SWBLI by Murphree et al. [13] found that transitional interactions resulting from a tripped boundary layer were similar to those with an upstream boundary layer that transitioned naturally.

The test cases will be referred to by their freestream condition (quiet/noisy) and insert type (smooth/tripped). Figure 5 contains sample schlieren images of the slice-ramp region from the three test cases. The mean-subtracted images were obtained by taking the difference between the raw images and the mean of the 200 surrounding images. In the raw images for all cases, a dark line indicates the

location of the shear layer over the separation bubble, with the separation shock visible above the shear layer. A portion of the nose-tip shock is visible in the upper-left corner. The noisy-smooth and quiet-tripped cases, which are both upstream-transitional, result in a significant reduction in separation length compared to the interaction-transitional quiet-smooth case. There is also a noticeable difference in freestream pressure fluctuation levels between the quiet and noisy schlieren images. Consistent with previous observations by Benitez [40], the increased freestream noise seems to interact with the shear layer and disturb its structure, with the shear layers in quiet flow appearing as cleaner lines in the schlieren; this is especially clear in the mean-subtracted images. Also, the nose-tip shock is only visible in the mean-subtracted image for noisy flow, as its motion is influenced by the freestream disturbances, whereas it is almost completely stationary in quiet flow.

3.1.1 Upstream boundary-layer characteristics

The pressure fluctuations in the upstream boundary layer, p'_{bl} , are measured with a PCB pressure transducer 10 mm upstream of the shock foot; the power spectral densities (PSDs) of these fluctuations are shown in Fig. 6. The measured pressure fluctuations are normalized by the boundary-layer edge pressure, p_e . The quiet-tripped and noisy-smooth boundary layers both contain higher levels of pressure fluctuations compared to the quiet-smooth case; however, while they are both transitional boundary layers, their frequency content do not align. The quiet-tripped boundary layer contains a significant peak around 250 kHz which indicates the presence of Mack's second-mode instability [34], while the power content of the noisy-smooth signal is more broadly distributed. The spike around 40 kHz present in all traces is attributed to sensor noise, not corresponding to any physical phenomenon. As previously mentioned, throughout one test in the BAM6QT, the freestream unit Reynolds number decreases as t increases; therefore, the boundary-layer thickness at the trip location increases with t , decreasing the boundary-layer trips' effective height as the test progresses. This effect is shown in Fig. 6 by plotting the PSDs for the upstream boundary layer at three different times throughout one quiet-tripped test. As t increases, there is a decrease in the magnitude of the second-mode instability, as well as a slight broadband decrease in power content. This provides an opportunity to study how varying degrees of boundary-layer transition and second-mode magnitude affect the SWBLI and separation bubble dynamics. The authors acknowledge that the differences in frequency content distribution between the quiet-tripped and noisy-smooth cases could introduce uncertainty in the comparison of quiet and noisy freestream effects; however, replicating the exact effects of noisy flow on the boundary layer was not possible in quiet flow.

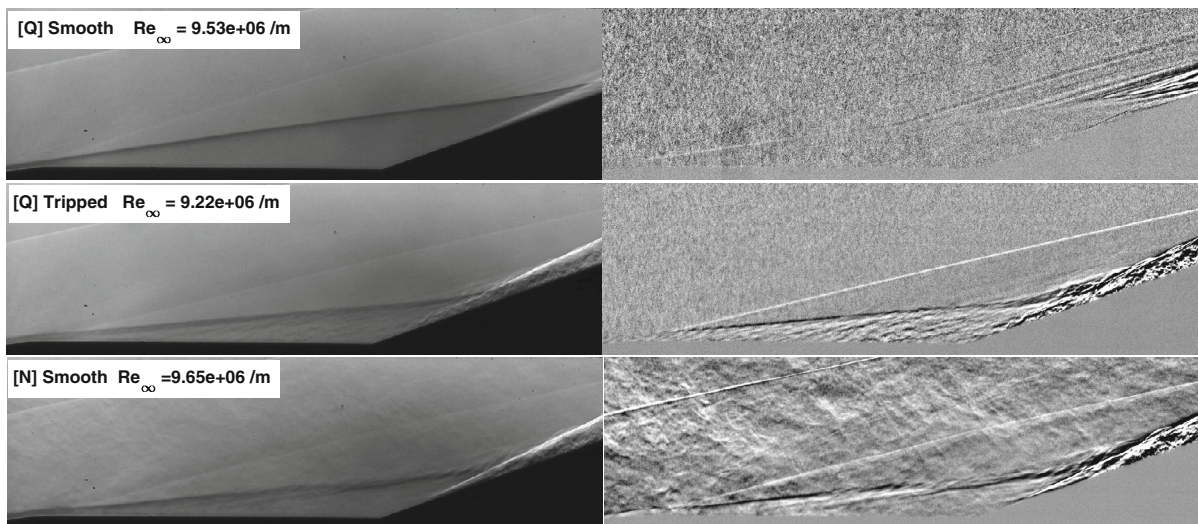


Fig. 5 Raw (left) and mean-subtracted (right) sample schlieren images of the slice-ramp region for the three test cases. Flow is from left to right. [Q] and [N] indicate quiet and noisy freestream flows, respectively

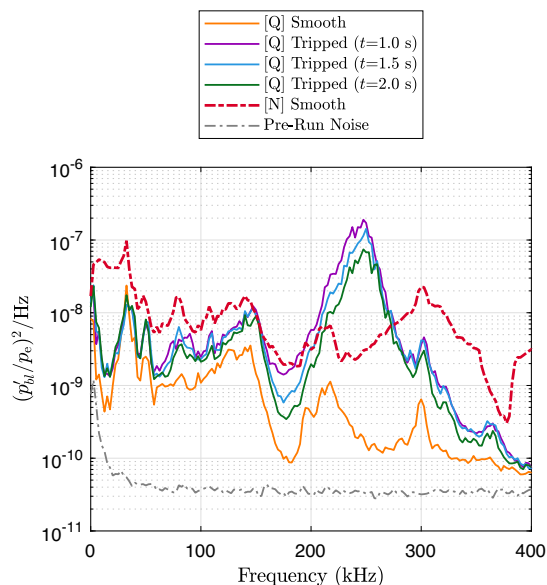


Fig. 6 Power spectral densities of pressure fluctuations in the boundary-layer upstream of the SWBLI

In this study, freestream noise and surface roughness elements are treated as mechanisms for modifying the incoming boundary-layer state, and the resulting SWBLI dynamics are interpreted as responding most directly to the state of boundary layer entering the interaction. With this distinction in mind, the following sections also examine how differences in freestream disturbance environment influence the unsteady behavior of the SWBLI when mean incoming boundary-layer properties are closely matched.

3.1.2 Reynolds number effects on quiet–tripped flow

Throughout a single test for the quiet–tripped case, the continual decrease in effective trip height causes significant changes to the upstream boundary layer and the separation bubble size. Figure 7 contains sample schlieren images of the separation bubble at various times throughout the quiet–tripped test, along with the freestream Reynolds number at that point in the test. As t increases, there is a visible fading of the second-mode rope waves in the boundary and shear layers and a significant increase in separation bubble length. Also, although the visible turbulence along the shear layer is less prominent as time increases, the reattached boundary layer appears turbulent throughout the entire test. In Fig. 8, the separation bubble length, L_{bub} , and the band-integrated root mean square (RMS) of the upstream pressure fluctuations are plotted with respect to the changing freestream Reynolds number and time, t , within the test. RMS values for the upstream boundary layer, $p'_{\text{bl,RMS}}$, are determined by integrating the PSD of the fluctuations over the 50–400 kHz frequency band and taking the square root of the result; they are reported as a percentage of the boundary-layer edge pressure. Throughout one test, a 4.1% decrease in Reynolds number causes a 42.4% decrease in the RMS of the pressure fluctuations in the tripped boundary layer, which in turn yields a 19.8% increase in the separation bubble length. Using these significant variations in upstream boundary-layer state, the effects of boundary-layer transition on SWBLI-induced unsteadiness can be studied by analyzing the behavior at different times throughout the quiet–tripped test.

For each configuration tested, the mean separation bubble length, the upstream boundary-layer thickness, and pressure

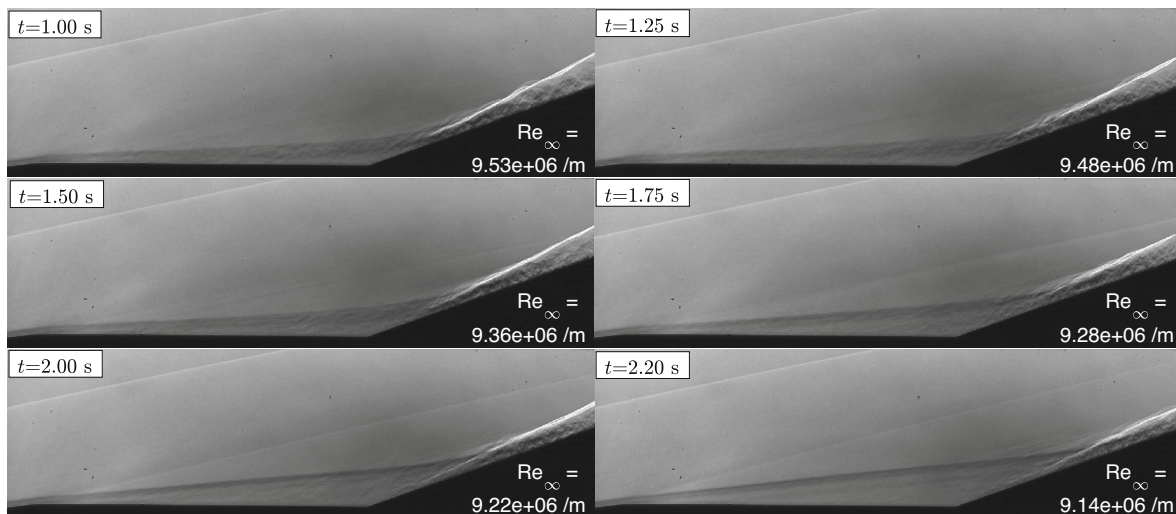


Fig. 7 Sample schlieren images of the separation bubble during a quiet test with boundary-layer trips, showing the variation of Reynolds number and separation length throughout a single test. Flow is from left to right

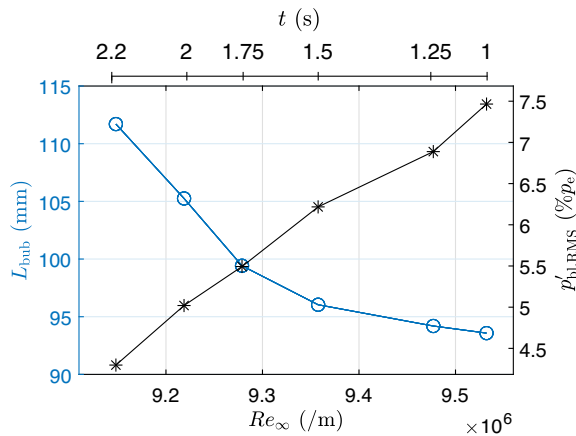


Fig. 8 Variation in separation bubble length and RMS of pressure fluctuations in the upstream boundary layer throughout a single test for the quiet-tripped case

fluctuation RMS values are presented in Table 1. For the purpose of comparing freestream noise effects on unsteady flow properties, it was desired that the mean flow properties were as similar as possible between quiet and noisy flow. This

is best achieved at $t = 2.0$ s in the quiet-tripped case; the percent differences between the mean flow properties of this case and the noisy-smooth case are also listed in Table 1. The percent differences for the mean flow properties measured for these two cases are all less than 6%; therefore, the direct comparison of unsteady flow behaviors for the quiet-tripped (at $t = 2.0$ s) and noisy-smooth cases will provide insight into the effects of freestream noise.

3.2 Unsteady flow properties

3.2.1 Separation height and pressure fluctuations

The unsteady behavior of the shock and separation bubble is characterized by tracking the motion of the shear layer according to the method described in Sect. 2.3. Figure 9 compares these separation bubble oscillations measured in the schlieren, h' , with the low-frequency pressure fluctuations inside the separation bubble, p'_{bub} . The height oscillations are reported as a percentage of the mean separation height, with a moving mean used for the quiet-tripped case in order to account for the gradually increasing mean separation height.

Table 1 Mean flow properties for each test configuration

	L_{bub} (mm)	δ_u (mm)	$p'_{\text{bl,RMS}}$ (% p_e)
Quiet-smooth	131.1	1.37	1.51
Quiet-tripped ($t = 1.0$ s)	93.6	1.41	7.47
Quiet-tripped ($t = 1.5$ s)	96.0	1.39	6.22
Quiet-tripped ($t = 2.0$ s)	105.3*	1.39 [†]	5.02 [‡]
Noisy-smooth	99.4*	1.35 [†]	4.74 [‡]

*5.8% diff.

[†]2.9% diff.

[‡]5.6% diff.

There is strong agreement between the height and pressure measurements, with a decrease in separation height corresponding to a decrease in pressure within the separation region. This is evidence that the shear-layer-tracking algorithm is able to accurately represent the low-frequency separation unsteadiness. In the cases with a transitional upstream boundary layer, the magnitudes of the shear-layer oscillations and pressure fluctuations within the bubble are generally greater than those in the quiet-smooth case, with one exception. Data from three different times within the quiet-tripped test are presented in Fig. 9, with the separation bubble unsteadiness behaving differently as time progresses. At $t = 1.0$ s, when the upstream boundary-layer fluctuations are at their highest level, the magnitude of the pressure fluctuations within the separation bubble is noticeably lower than all other cases, including the quiet case with the smooth insert. Although the quiet-tripped and noisy-smooth cases both have transitional incoming boundary layers, the oscillation magnitudes in the noisy-smooth case are noticeably higher.

3.2.2 Power spectral density

The fluctuations shown in Fig. 9 are analyzed using power spectral density analyses to investigate the effect of transition and freestream noise on the frequency content of the unsteadiness. Figure 10 shows the PSDs of the static pressure fluctuations measured by the Kulite pressure transducer within the separation bubble. In Fig. 10b, the PSDs are pre-multiplied by frequency and plotted with respect to Strouhal number. Interestingly, for the quiet-tripped test, as t increases and the upstream boundary-layer fluctuations decrease, the power content of the low-frequency pressure fluctuations in the bubble generally increases. In fact, the quiet-tripped flow at $t = 1.0$ s causes significantly lower pressure fluctuation levels in the separation bubble than even the quiet flow with the smooth insert. This is an indication that, as boundary-layer transition progresses and second-mode magnitudes increase, the additional forcing on the system does not necessarily increase the unsteady behavior; in fact, it may be suppressing it. Additionally, the decrease in separation length resulting from the increase in upstream fluctuations may be partly responsible; Sansica et al. [19] observed that a larger separation bubble had a stronger low-frequency response to forcing. The noisy flow yields low-frequency static pressure fluctuations that are significantly higher in magnitude than any of the quiet flow cases, suggesting that the additional turbulent pressure fluctuations introduced by the freestream may be exciting or magnifying the SWBLI-induced unsteadiness.

The PSDs of the separation bubble height fluctuations are plotted in Fig. 11, with respect to both frequency and Strouhal number. In all cases, there is evidence of low-frequency

separation unsteadiness, with the highest power content corresponding to frequencies below 2 kHz. The quiet-smooth case, which is only transitional near reattachment, has broadband unsteadiness at frequencies below 1.5 kHz, with slight peaks at 0.3 kHz and 1 kHz and a sharp decline around 2 kHz; this is consistent with previous studies on the cone-slice-ramp under quiet Mach-6 flow [32, 33]. In the pre-multiplied spectra, this appears as a small peak around $Sr_L = 0.04$ and a larger peak around $Sr_L = 0.15$. The quiet-tripped case exhibits greater unsteadiness at $Sr_L = 0.04$ than the quiet-smooth case, but in the quiet-tripped case, this peak grows with increasing time and therefore decreasing upstream boundary-layer fluctuations. The PSDs corresponding to the quiet-tripped case do not vary with time as dramatically as in Fig. 10, but there is still a significant shift as time progresses. The shift in frequency content for the tripped case appears hinged around the 2-kHz point (or $Sr_L \approx 0.2$), with increasing time—and decreasing second-mode magnitudes—causing a decrease in power content above this point, but an increase in power content below. Specifically, as second-mode magnitude decreases, a peak develops at $Sr_L \approx 0.035$. While it appears that increasing pressure fluctuations in the upstream boundary layer can excite low-frequency unsteadiness in quiet flow, there appears to be a point at which that trend reverses. From both Figs. 10 and 11, it is clear that the magnitude of the second-mode instability and the degree of transition in the upstream boundary layer affect the unsteadiness magnitudes and frequency content in quiet flow. Consistent with what was previously observed in Fig. 9, the magnitude of the unsteadiness is significantly greater for the noisy flow case compared to all quiet cases. Its lowest-frequency peak occurs at $Sr_L \approx 0.045$, with another clear peak at $Sr_L \approx 0.15$. These peaks also appear in the pressure fluctuation measurements shown in Fig. 10b.

All cases exhibit low-frequency unsteadiness in the range of $Sr_L = 0.025$ – 0.05 , values which are often associated with large-scale bubble breathing for both transitional and turbulent SWBLI [3, 17, 18, 20]. This result is notable as the low-frequency unsteadiness on this geometry is a large-scale flapping motion, rather than a breathing one. This implies that the low-frequency unsteadiness at $Sr_L = 0.025$ – 0.05 is not exclusive to the bubble breathing motion. One possible explanation for this is that the flapping originates from the same mechanism as the breathing motion in other fully separated flows. According to the theory of fluid entrainment as a downstream mechanism [11], the bubble breathing arises because the bubble grows and shrinks based on a cycle of mass entrainment and release. For this interaction, it is possible that this downstream mechanism is present, but the range of motion of the separation bubble is limited by the fixed separation point, resulting in bubble growth in the form of flapping rather than breathing.

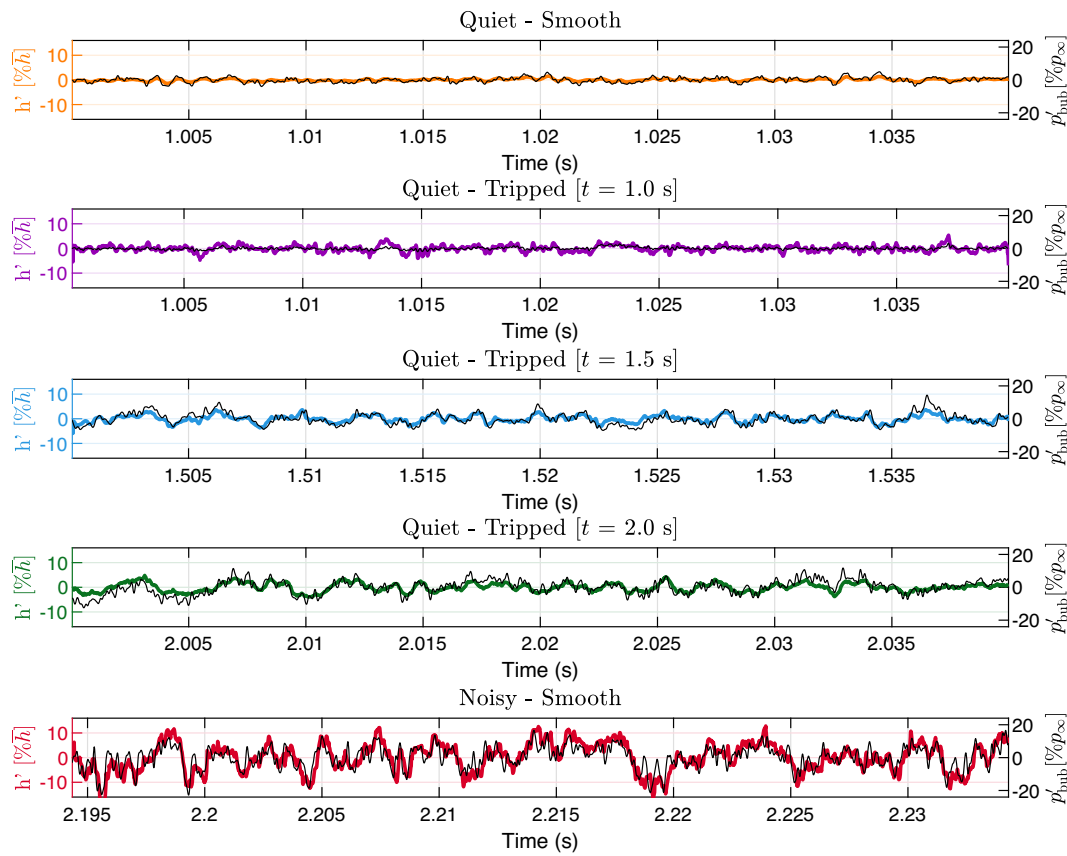


Fig. 9 Time traces of separation bubble height fluctuations tracked in the schlieren alongside the static pressure fluctuations measured inside the separation bubble. The pressure data have been low-pass filtered at 1 kHz and smoothed using a moving-average filter to reveal only the low-frequency fluctuations

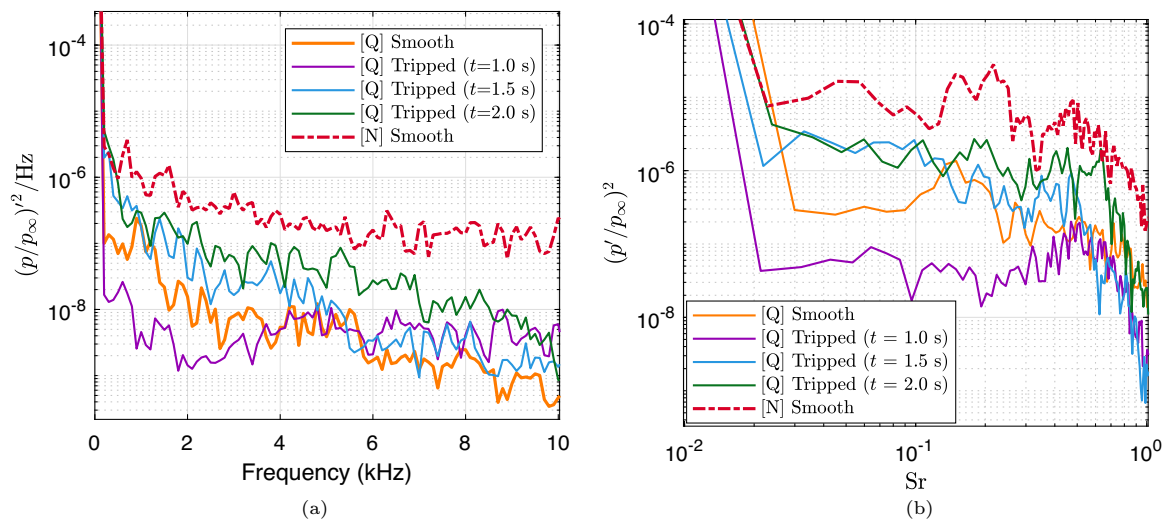


Fig. 10 Power spectral densities of the static pressure fluctuations measured by a Kulite pressure transducer within the separation bubble, plotted with respect to **a** frequency and **b** Strouhal number with a pre-multiplied y-axis

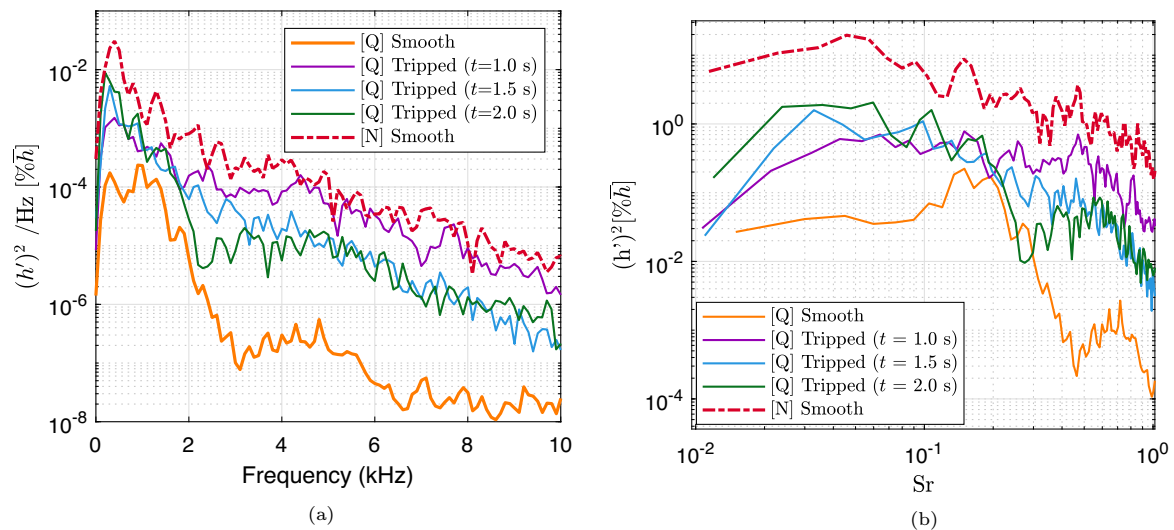


Fig. 11 Power spectral densities of the separation height fluctuations plotted with respect to **a** frequency and **b** Strouhal number with a pre-multiplied y-axis

The differences in trends between Figs. 10 and 11 for the quiet–tripped case could be explained in part by the normalization applied. The height fluctuations were normalized by mean height, which saw a dramatic increase as time increased (see Fig. 7), but the pressure fluctuations were normalized by the freestream pressure, which decreased only slightly throughout the test. This would explain why the difference in the separation height oscillations at $t = 1.0$ s is not as dramatic as it is for the static pressure fluctuations. For this reason, subsequent RMS values are presented with and without this normalization applied.

3.2.3 Root mean square of low-frequency unsteadiness

Figure 12 compares RMS values of the upstream boundary-layer pressure fluctuations with RMS values for the low-frequency separation bubble unsteadiness. The RMS values for the low-frequency fluctuations in separation bubble height, h'_{RMS} , and static pressure in the bubble, $p'_{\text{bub,RMS}}$, are determined by integrating the PSDs over the 0.2–5 kHz band and taking the square root of the result. The integration band for the upstream boundary-layer RMS values is 50–400 kHz. The RMS values presented were calculated at six time steps throughout a test for the quiet–smooth, quiet–tripped, and noisy–smooth cases. For the cases without boundary-layer trips, it is evident that RMS values from quiet flow (asterisks) contain significantly less spread than the results from noisy flow (diamonds). Figure 12 shows that higher levels of pressure fluctuations in the boundary-layer upstream of the SBWLI do not always correspond to greater unsteadiness magnitudes in the separation bubble. In quiet flow, as boundary-layer transition progresses and second-mode amplitudes increase, the RMS magnitudes

for the unsteadiness reach a peak value, then decrease as the boundary-layer RMS magnitudes continue to increase. Experiments on a transitional hypersonic cylinder interaction by Murphree et al. [13] found a similar relationship, with unsteadiness decreasing as the degree of boundary-layer transition increased. The RMS values for the static pressure fluctuations in the bubble decrease beyond those of the quiet–smooth case; at the highest second-mode magnitudes, the low-frequency oscillatory pressure loads are weaker than those resulting from the laminar upstream boundary layer.

Confirming the observations in Figs. 10 and 11, the separation unsteadiness RMS values from noisy flow are significantly greater than those from quiet flow, for both the physical motion of the bubble and the pressure fluctuations within it. For the non-normalized data, the RMS values of the unsteadiness in noisy flow are approximately double those measured in quiet flow with similar boundary-layer RMS values. In the normalized data, the lowest RMS values measured for noisy flow are 56% and 173% greater than the highest values measured in quiet flow, for the pressure and height fluctuations, respectively. This significant increase in the unsteadiness RMS magnitudes in noisy flow suggests that freestream pressure fluctuations are capable of exciting the mechanisms responsible for SBWLI-induced unsteadiness.

3.2.4 Magnitude-squared coherence and phase shift

The magnitude-squared coherence, C_{xy} , of the intensity of each pixel in a set of 3040 schlieren images is calculated with respect to a probe pixel to determine whether there are any coupled structures in the flow. Additionally, the phase angle, ϕ , of the cross-power spectral density, P_{xy} , is determined for each pixel in relation to the probe pixel in order to determine lead/lag relationships between the structures. The

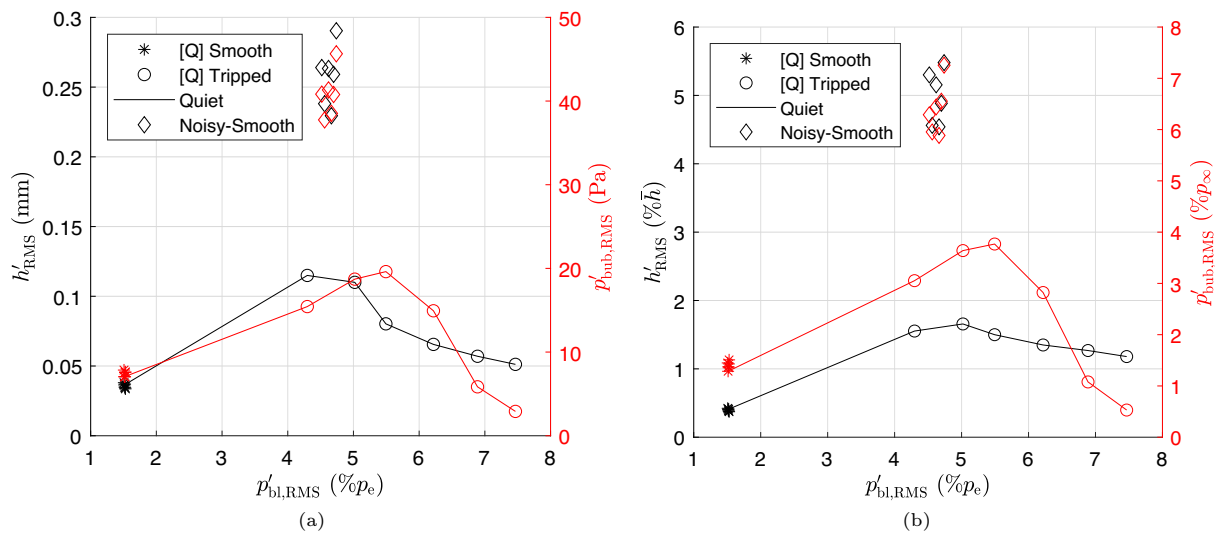


Fig. 12 Effect of upstream boundary-layer fluctuations on low-frequency band-integrated RMS values for separation bubble height and static pressure fluctuations, **a** without normalization, and **b** with normalization

cross-power spectral densities are calculated so that a positive ϕ corresponds to the pixel of interest leading the probe pixel. For frequency bands of interest, the mean phase angle for each pixel, $\bar{\phi}$, is determined using a weighted average:

$$\bar{\phi} = \frac{\sum_f (C_{xy}(f) \times \phi(f))}{\sum_f C_{xy}(f)}. \quad (1)$$

Both analyses have a frequency resolution of 74.3 Hz.

In a method similar to the one used in [17], the magnitude-squared coherence values are plotted over the phase angle values as a transparency map. Using this method, only the phase angles for areas of high coherence show through in order to reduce noise from non-coherent regions. Coherence values below 0.3 are completely opaque, and coherence values above 0.7 are entirely transparent, with a linear gradient between these two values; the model surface is also opaque. In addition, a Weiner filter has been applied to reduce the influence of noise from the camera sensor. Figure 13 shows the coherence/phase plots for a low frequency band ($Sr_L \approx 0.03$ – 0.05) and a mid-frequency band ($Sr_L \approx 0.14$ – 0.155) with respect to a probe pixel in the shear layer. In general, the side of the shear layer that the probe is on will be in phase with the pixel ($\phi = 0$), and the opposite side will be out of phase ($\phi = \pi$), as the movement of the shear layer that darkens the probe side will lighten the opposite side, and vice versa. For the quiet cases, the probe pixel is on the top side of the shear layer, and for the noisy case it is on the bottom side; this was an unintended consequence of probe selection, and does not affect the interpretation of the results.

For all cases, it is evident that the separation shock motion is coherent with the shear layer for both frequency bands.

In the quiet-smooth case, there is weak coherence with the unsteady downstream shocklets; however, the reattachment region is not entirely within the field of view. There is much higher coherence with the reattachment region for the quiet-tripped and noisy-smooth cases, where the shear layer becomes turbulent as it interacts with the ramp (see Fig. 5). For the quiet-tripped and noisy-smooth flows in the mid-frequency band (Fig. 13d and f), the most downstream region of the shear layer appears to lead the probe pixel by approximately $\pi/4$, as evidenced by the blue region at reattachment within the in-phase side of the shear layer. This implies that motion of the shear layer near reattachment proceeds the large-scale shock and separation flapping, which suggests a downstream mechanism driving the unsteadiness. This result is consistent with the findings of Wu and Martin [10] for 2D turbulent interactions and supports the theory by Sansica et al. [20] that the unsteadiness in transitional SWBLI is related to turbulence near reattachment.

Compared to the results from quiet flow, the coherence/phase plots for the noisy-smooth case contain broader areas of high coherence around the separation shock and shear layer. This is indicative of the higher-magnitude unsteadiness in noisy flow, as the shock and shear layer are influencing a wider range of pixels as they oscillate; it is consistent with the behavior observed in Fig. 9. Besides noisy flow having freestream pressure fluctuations that are two orders of magnitude greater than those in quiet flow, the only other main difference between the noisy-smooth and quiet-tripped cases is the distribution of frequency content in the upstream boundary layer (see Fig. 6). Although the distribution of the frequency content in the transitional boundary layer could potentially influence the unsteady behavior, it is more likely that the greater unsteadiness amplitudes in noisy

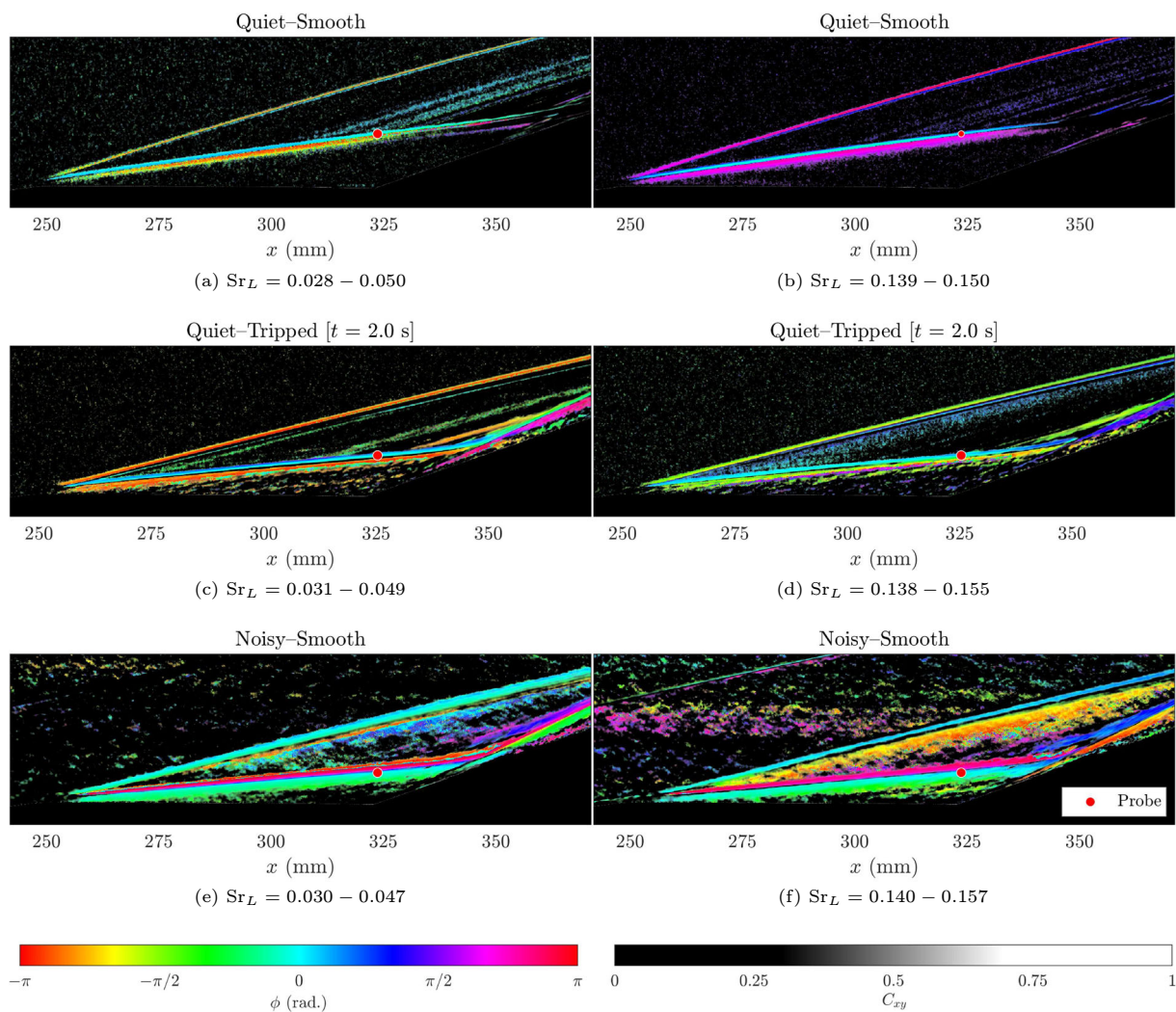


Fig. 13 Phase shift with respect to a probe pixel in the shear layer for high-coherence regions in the schlieren for the low-frequency (left column) and mid-frequency (right column) bands

flow are related to the significantly higher levels of forcing from the freestream.

There is also evidence of the influence of wind-tunnel noise in the mid-frequency band for the noisy-smooth flow (Fig. 13f), where there is coherence with the nose-tip shock in the upper left corner. Also, there is a region in the upstream flow that exhibits both high coherence and a consistent phase relative to the probe pixel (the pink/purple region between the nose-tip shock and the separation shock). It is unknown how exactly this region of upstream flow is related to the shear-layer motion, but the path of this high-coherence region suggests that the fluctuations within it may interact with the reattachment region as they move downstream. Because large separated bubbles such as these are more sensitive to downstream mechanisms [19], and because the modulation of entrainment rate may be responsible for unsteady SWBLI behaviors [12], the authors propose that the interaction of

freestream fluctuations with the downstream shear layer and reattachment regions may be exciting the downstream mechanism and causing the higher levels of unsteadiness observed in noisy flow. This would imply that measurements of transitional SWBLI-induced unsteadiness in noisy wind tunnels could be artificially inflated by freestream noise effects.

In order to further study the effects of freestream pressure fluctuations, a coherence/phase analysis is performed for a probe pixel within the nose-tip shock, since previous work in the BAM6QT [41] has shown that the motion of this shock is sensitive to freestream pressure fluctuations and therefore can often be representative of their influence. As the nose-tip shock in quiet flow is almost completely stationary, only one quiet flow case is analyzed in this manner to provide a comparison against the result from noisy flow. The coherence/phase plots for the nose-tip shock are shown in Fig. 14 for the same two frequency bands used in Fig. 13. Firstly,

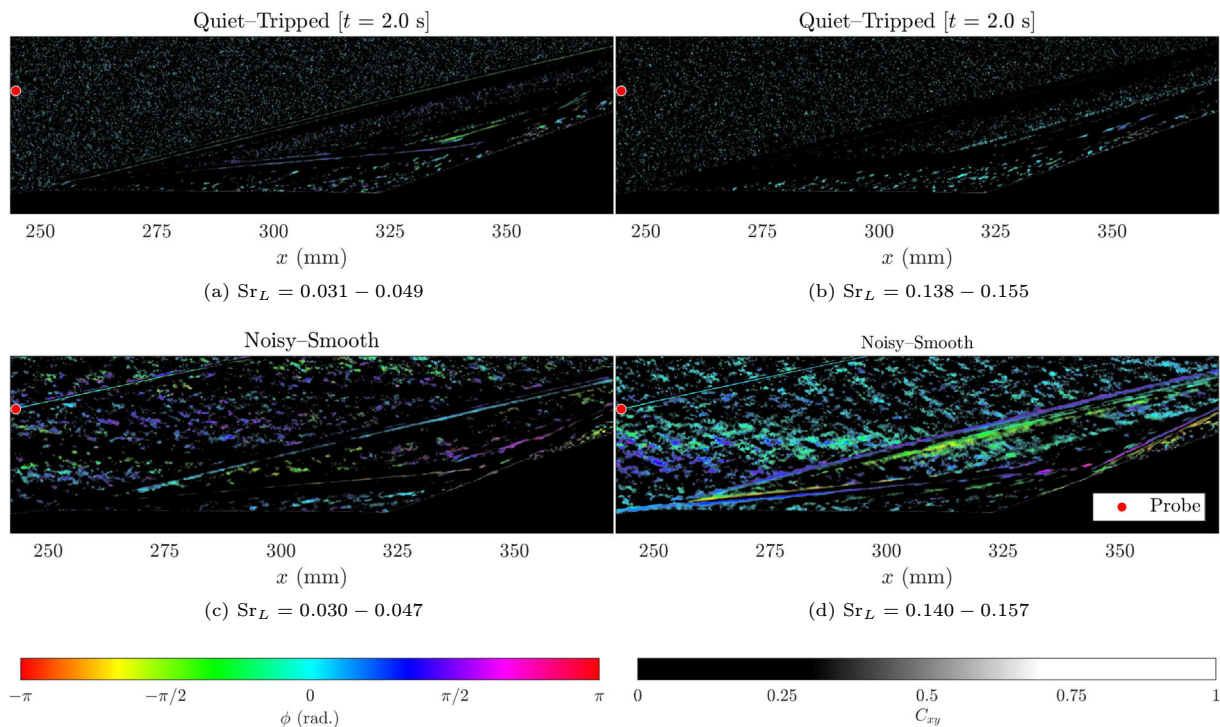


Fig. 14 Phase shift with respect to a probe pixel in the nose-tip shock for high-coherence regions in the schlieren for the low-frequency (left column) and mid-frequency (right column) bands

for the noisy freestream case, the nose-tip shock is highly coherent and in phase with itself in both frequency bands, indicating that the shock does contain unsteadiness at the frequencies within those bands. Also, in noisy flow, the probe in the nose-tip shock exhibits coherence with the motion of the separation shock and shear layer which is not present in quiet flow; the coherence in noisy flow is especially strong for the mid-frequency band. In this band, the areas of the separation shock and shear layer that have the highest coherence with the probe pixel are generally out of phase with the nose-tip shock by $\phi = \pi/2$. The region of high coherence upstream of the separation shock in Fig. 13f is also present in Fig. 14d and generally appears in phase with the nose-tip shock. The coherence values in the low-frequency band are generally too low to draw any meaningful conclusions from, although there exist a couple of higher-coherence regions in the separation shock in noisy flow. It is important to note that the displacement of the nose-tip shock is likely not the only way that freestream pressure fluctuations influence the interaction, so not all freestream noise effects will be visible in these coherence images.

4 Conclusion

The low-frequency unsteady dynamics of a three-dimensional shock-wave/boundary-layer interaction is stud-

ied in a Mach-6 wind tunnel capable of low-disturbance and conventional-noise flow. The increased freestream pressure fluctuations in noisy flow result in a transitional boundary-layer upstream of the SWBLI; boundary-layer trips are used in quiet flow to create a transitional boundary layer by exciting the second-mode instability. Accordingly, the observed differences in SWBLI unsteadiness are interpreted primarily in terms of the state of the boundary layer entering the interaction, with freestream noise and surface roughness acting as disturbance mechanisms that modify this state. For conditions resulting in similar upstream boundary-layer characteristics between quiet and noisy flow, the influence of the additional external forcing by freestream pressure fluctuations is examined.

The SWBLI-induced unsteadiness is studied in quiet flow for a range of effective trip heights and, therefore, varying degrees of upstream boundary-layer transition. In quiet flow, as the pressure fluctuation RMS values in the upstream boundary layer increase, there is an increase then decrease in the RMS magnitudes of the low-frequency unsteadiness—measured using both separation bubble height and interior pressure fluctuations. This behavior suggests that low-frequency SWBLI-induced unsteadiness is highly sensitive to the degree of transition in the upstream boundary layer, with greater second-mode amplitudes not necessarily yielding greater unsteadiness magnitudes. In noisy flow with a transitional upstream boundary layer, the resulting

unsteadiness magnitudes are significantly greater than those measured in quiet flow. For one quiet flow case with a tripped boundary layer, its separation bubble length, and upstream boundary-layer height and pressure fluctuation RMS are all within 6% of those measured in the noisy flow case, yet the magnitudes of the unsteadiness in noisy flow are approximately double those measured in the quiet flow case. This result suggests that wind-tunnel noise may affect experimental measurements of unsteady behavior in transitional SWBLI due to the system's sensitivity to external disturbances. The authors propose that this could be due to the interaction of freestream pressure fluctuations with the shear layer, affecting the mass entrainment rates. As the highly three-dimensional nature of this SWBLI undoubtedly affects its dynamic behavior, further studies may be necessary to discern if the findings from this study apply to canonical 2D geometries. In addition, further studies are needed before drawing conclusions on the effect of wind-tunnel noise on turbulent SWBLI, as the turbulent shear layer and smaller separation scale may affect the system's receptivity to disturbances from the freestream.

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Declarations

Conflict of interest The authors have no conflict of interest to declare.

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